

ADR IDENTIFICATION BASICS

Patient Case Analysis for Suspected Adverse Drug Reactions

VirtualWorks Lab Internship - Pharmacovigilance Task

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Educational case-analysis submission

1. Aim and Objective

Aim: To identify possible adverse drug reactions by reviewing patient symptoms, medicines taken, timing of the reaction, and other possible causes.

Objective: To decide whether a reported event is likely to be drug-related and to identify the most likely suspected medicine using a simple causality-based approach.

2. How I Assessed Each Case

For each case I first looked at what happened to the patient and when it happened. I then compared the event with the known adverse effects of the medicine. I also checked whether the patient improved after stopping or reducing the drug, and whether another condition, medicine, infection, food-related problem, or change in diet could explain the same symptoms.

Point checked	What I looked for
Reaction	What symptoms or clinical signs were reported?
Medicine history	Which drug was started, stopped or changed recently? Were there other medicines?
Time relationship	Did the reaction start after the medicine was started or the dose was increased?
Known adverse effect	Is the reaction already known to occur with this medicine or drug class?
Dechallenge	Did the problem improve after the drug was stopped or the dose was reduced?
Other causes	Is there any reasonable non-drug explanation?
Conclusion	Which drug is most suspicious and how strong is the relationship?

3. Simple WHO-UMC Causality Guide

I used the WHO-UMC terminology only as a practical guide for the cases below. In real pharmacovigilance work, causality depends on the amount and quality of information available in the individual case report.

Certain: Strong time relationship, no better explanation, convincing response to withdrawal, and sufficiently clear evidence.

Probable / Likely: Reasonable timing, the reaction is consistent with the medicine, other causes are less likely, and dechallenge may support the link.

Possible: The drug relationship is reasonable, but another cause could also explain the event or the information is incomplete.

Unlikely: The timing does not fit well or another explanation is more convincing.

Unassessable: There is not enough reliable information to make a causality judgment.

4. Patient Case Analysis

The following are fictional practice cases prepared only for internship learning. They are not real patient records.

Case 1 - Dry cough after starting enalapril

Patient	46-year-old man with hypertension
Medication	Enalapril 5 mg once daily
Case details	The patient started enalapril and developed a persistent dry cough about 10 days later. He did not report fever, sputum or other features suggesting a respiratory infection. The cough gradually settled after enalapril was stopped and another antihypertensive was used.
Potential ADR	ACE inhibitor-associated dry cough
Suspected drug	Enalapril
Causality	Probable / Likely

My assessment: The timing fits because the cough started after treatment began. Dry, non-productive cough is a well-known effect of ACE inhibitors. There was no clear evidence of infection in the case, and the cough improved after withdrawal. For these reasons I would consider enalapril the main suspected drug.

Case 2 - Ankle swelling after amlodipine dose increase

Patient	58-year-old woman being treated for hypertension
Medication	Amlodipine increased from 5 mg to 10 mg once daily
Case details	About two weeks after the dose was increased,

	the patient noticed swelling around both ankles. She did not have breathlessness or orthopnea. The swelling became less after the dose was reduced.
Potential ADR	Peripheral edema
Suspected drug	Amlodipine
Causality	Probable / Likely

My assessment: The reaction appeared after a dose increase and improved when the dose was lowered. Ankle edema is a recognized and dose-related adverse effect of amlodipine. Heart failure is an important alternative cause of edema, but the case does not describe symptoms that strongly suggest it.

Case 3 - Gastrointestinal symptoms after starting metformin

Patient	52-year-old man newly diagnosed with type 2 diabetes
Medication	Metformin 500 mg twice daily
Case details	Within three days of starting metformin, the patient developed loose stools, nausea and abdominal discomfort. He had no fever or similar symptoms before treatment. The symptoms improved when metformin was temporarily withheld and were milder when it was restarted gradually with meals.
Potential ADR	Metformin-related gastrointestinal intolerance
Suspected drug	Metformin
Causality	Probable / Likely

My assessment: There is a close time relationship and the symptoms are common with metformin, especially at the beginning of treatment. Improvement after temporary withdrawal supports the relationship. The case does not strongly suggest an infectious cause.

Case 4 - Skin rash during amoxicillin treatment

Patient	24-year-old woman treated for pharyngitis
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Medication	Amoxicillin 500 mg three times daily
Case details	A generalized itchy maculopapular rash appeared on the fourth day of treatment. She had no breathing difficulty, facial swelling or mucosal involvement. The rash gradually faded after amoxicillin was stopped.
Potential ADR	Suspected drug-related maculopapular rash
Suspected drug	Amoxicillin
Causality	Possible

My assessment: The timing is compatible and amoxicillin can cause cutaneous reactions. However, some viral illnesses can also cause a similar rash, and the original case does not give enough information to completely exclude a viral exanthem. I would therefore classify this relationship as possible rather than definite.

Case 5 - Gastric irritation after ibuprofen use

Patient	35-year-old man taking medicine for back pain
Medication	Ibuprofen 400 mg three times daily for five days
Case details	After several days of ibuprofen use, the patient developed burning epigastric pain and dyspepsia, particularly when he took the tablets without food. He reported no vomiting of blood or black stools. Symptoms improved after ibuprofen was stopped and symptomatic treatment was given.
Potential ADR	NSAID-associated dyspepsia / gastric irritation
Suspected drug	Ibuprofen
Causality	Probable / Likely

My assessment: The onset followed repeated NSAID use and the symptoms are consistent with the known gastrointestinal effects of ibuprofen. The improvement after stopping the medicine also supports the link. Other causes of dyspepsia are possible, but they are less strongly supported by the information given.

Case 6 - Hypoglycemia after insulin dose increase

Patient	61-year-old woman with diabetes using basal insulin
Medication	Insulin glargine; dose recently increased
Case details	During the night the patient developed sweating, tremor, palpitations and confusion. Capillary blood glucose was 48 mg/dL. The symptoms resolved after glucose was given. The insulin dose had recently been increased, and the patient had also eaten a smaller evening meal than usual.
Potential ADR	Hypoglycemia
Suspected drug	Insulin glargine
Causality	Probable / Likely

My assessment: The symptoms were confirmed by a low glucose reading and occurred after an insulin dose increase. The smaller meal probably contributed to the event, so it is not the drug dose alone, but insulin remains the main suspected medicine because its glucose-lowering action directly explains the reaction.

5. Summary of Findings

Case	Suspected drug	Potential ADR	Causality
1	Enalapril	ACE inhibitor-associated dry cough	Probable / Likely
2	Amlodipine	Peripheral edema	Probable / Likely
3	Metformin	Metformin-related gastrointestinal intolerance	Probable / Likely
4	Amoxicillin	Suspected drug-related maculopapular rash	Possible
5	Ibuprofen	NSAID-associated dyspepsia / gastric irritation	Probable / Likely

Case	Suspected drug	Potential ADR	Causality
6	Insulin glargine	Hypoglycemia	Probable / Likely

6. Conclusion

From these cases, I understood that identifying an ADR is not based only on the fact that a symptom happened after a medicine was taken. The time of onset, known adverse effects, response after stopping or reducing the medicine, dose changes and other possible causes all need to be considered together. In most of the cases above, the suspected drug had a reasonable temporal relationship and the clinical pattern matched a known drug effect. The amoxicillin rash was kept as a possible reaction because an alternative explanation could not be fully ruled out from the available information.

7. References

- World Health Organization - Uppsala Monitoring Centre. The use of the WHO-UMC system for standardised case causality assessment.
- Pharmacovigilance Programme of India (PvPI). Guidance material for spontaneous adverse drug reaction reporting and causality assessment.
- British National Formulary / standard drug information sources for commonly recognized adverse effects of medicines used in the practice cases.

Note: All patient scenarios in this report are fictional and are included only to demonstrate ADR identification for an educational internship task. No real patient information has been used.